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Multi-Criteria Decision Analysis for the Evaluation of Nuclear Reactor Designs

Tesi di Laurea Magistrale in
Nuclear Engineering - Ingegneria Nucleare

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Abstract

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Abstract in lingua italiana

Qui va l'Abstract in lingua italiana della tesi seguito dalla lista di parole chiave.

Parole chiave: qui, vanno, le parole chiave, della tesi

Contents

Introduction

A brief History of Nuclear Power, from its origins to present day

The 1930s and 1940s witnessed unprecedented acceleration in nuclear science. Following James Chadwick's 1932 discovery of the neutron [4], Otto Hahn and Fritz Strassmann demonstrated nuclear fission in 1938 by producing barium through uranium neutron bombardment [4]. This breakthrough enabled Enrico Fermi's team to construct Chicago Pile-1 in 1942 - the first artificial nuclear reactor capable of sustaining and controlling a chain reaction [4] - just four years after the fission discovery.

While initial advancements were primarily motivated by military applications during World War II, culminating in the atomic bombings of Hiroshima (August 6, 1945) and Nagasaki (August 9, 1945), the scientific community quickly recognized nuclear technology's peaceful potential. President Eisenhower's 1953 "Atoms for Peace" address to the United Nations [4] established the foundation for international nuclear cooperation, leading to the 1957 creation of both the International Atomic Energy Agency (IAEA) and the European Atomic Energy Community (EURATOM). These institutions were tasked with promoting peaceful nuclear applications while serving as independent authorities for safety regulation and non-proliferation oversight.

The 1950s and 1960s marked a period of rapid nuclear reactor development, driven by dual civilian and military imperatives. This era saw several key milestones in nuclear power generation: the Experimental Breeder Reactor I (EBR-I) in Idaho (USA) first produced electricity in 1951 (powering four light bulbs) [4], followed by the Obninsk Nuclear Power Plant (USSR) in 1954, which became the first to feed approximately 5 MWe into a local grid [4]. The world's first commercial-scale nuclear power station, Calder Hall-1 (UK), began operation in 1956, connecting to the national grid with an initial capacity of 50 MWe [4].

Parallel technological diversification occurred during this period. The United Kingdom developed gas-cooled Magnox reactors, optimized for both electricity production and plutonium generation. Concurrently, the Soviet Union initiated development of the graphite-moderated, water-cooled RBMK design through experimental work at Obninsk. Canada's nuclear program focused on pressurized heavy water reactor (PHWR) technology, culminating in the 1962 Rolphton demonstration reactor and the first commercial CANDU reactor at Douglas Point in 1968 [4].

In reactor cooling technologies, the United States pioneered multiple approaches: the EBR-I employed liquid metal (NaK) cooling, while the naval propulsion program led to light water reactor (LWR) development with the 1954 launch of USS Nautilus. By 1957, both Boiling Water Reactor (BWR) and Pressurized Water Reactor (PWR) designs were adapted for civilian applications, with the Shippingport Reactor (Pennsylvania) becoming the first full-scale civilian PWR connected to the grid. The Soviet VVER ("Water-Water Energetic Reactor") program advanced simultaneously, deploying the 210 MWe VVER-210 at Novovoronezh Nuclear Power Plant in 1964 [4].

The 1960s and 1970s witnessed significant global expansion of nuclear power generation. During this period, numerous countries commissioned their first nuclear reactors: Argentina (Atucha-1, 1974), Armenia (Armenian-1, 1976), Belgium (BR-3, 1962), Bulgaria (1974), Italy (Latina, 1963), Japan (1963), Sweden (1964), Switzerland (1969), and Slovakia (1972) [4].

The 1973 oil crisis accelerated nuclear adoption as an energy security strategy. France implemented the Messmer Plan, an ambitious program that resulted in the construction of 43 reactors between 1977 and 1986 [4]. Similarly, Japan significantly expanded its nuclear capacity to reduce dependence on imported oil, establishing nuclear energy as a cornerstone of its national energy strategy [4].

The nuclear expansion continued through the 1980s until the 1986 Chernobyl disaster, which constituted a critical inflection point for global nuclear development. The accident prompted widespread safety reviews and regulatory updates, significantly slowing deployment rates [4]. Public opposition intensified particularly in Western Europe, leading several countries to halt or reduce their nuclear programs. Italy's 1987 referendum resulted in the complete phase-out of nuclear power, while other European nations experienced similar setbacks [4].

By the 1990s, multiple countries had suspended nuclear deployment entirely, including Germany and Switzerland. France, previously the most active builder, connected only 10 reactors between 1990-1999, while the United States and United Kingdom deployed 4 and 1 reactors respectively during the same period [4]. Italy completed its nuclear phase-out by 1990 with the permanent shutdown of all operating reactors. Following the post-Soviet economic transition, Russia resumed its nuclear program and has since maintained continuous expansion of its reactor fleet [4].

A nuclear renaissance emerged in the early 2000s, driven by concerns over climate change, energy security, and rising fossil fuel prices. This period saw the development of advanced reactor designs including the European Pressurized Reactor (EPR) and AP1000 [4]. However, the 2011 Fukushima Daiichi accident constituted another significant setback for nuclear expansion in Western nations. Projects faced substantial delays, cost overruns, and public opposition, with France's Flamanville-3 EPR serving as a particularly illustrative case: experiencing budget increases to eight times original estimates and construction timelines tripled from initial projections [4]. Germany accelerated its nuclear phase-out policy, while Italy reaffirmed its anti-nuclear position through a popular referendum [4].

Contrastingly, Eastern nations pursued markedly different nuclear trajectories. China, having initiated its nuclear program in the 1990s with Qinshan-1 (connected in 1991), has since maintained aggressive expansion. Over the past three decades, China has brought 56 additional reactors online and currently has 29 under construction, positioning itself to surpass both France and the United States in total reactor capacity within coming decades [4]. This expansion reflects a strategic commitment to nuclear energy as part of China's broader energy and climate objectives.

The COVID-19 pandemic in 2020 exposed critical vulnerabilities in global economic systems, while Russia's 2022 invasion of Ukraine further revealed Europe's energy security dependencies [4]. Concurrently, decarbonization imperatives to combat climate change have prompted Western nations to reconsider nuclear energy's dual role: providing low-carbon, reliable baseload power while enhancing energy independence and system resilience [10].

This strategic reassessment is evident in recent policy initiatives. The European Union's classification of nuclear power as a sustainable investment under its taxonomy for climate mitigation [11], alongside programs like the U.S. Department of Energy's Advanced Reactor Demonstration Program [4], demonstrates renewed interest in nuclear technologies. The European energy sector is consequently undergoing a significant transformation, balancing decarbonization targets with energy security requirements through diversified generation portfolios that include advanced nuclear solutions [10].

While the political decision to include nuclear power is fundamentally a strategic choice, selecting specific reactor designs involves complex trade-offs across technical, economic, regulatory, social, environmental, and geopolitical dimensions [4]. National energy needs vary significantly: some countries require large baseload capacity to reduce import dependence, while others prioritize smaller, more flexible solutions to integrate with existing renewable systems [4].

The current nuclear landscape presents particular challenges for decision-makers, offering both established Generation III+ designs and emerging Small Modular Reactors (SMRs) as potential options. Given that nuclear commitments represent decade-long investments with 60-year operational lifespans [4] and significant financial implications, and considering the multiple influencing factors like grid infrastructure, public acceptance, and supply chain considerations, policymakers require structured decision-support tools to evaluate these complex trade-offs. This study therefore employs Multi-Criteria Decision Analysis (MCDA), specifically the Multi-Attribute Value Theory (MAVT) approach, presented in the following paragraphs.

MCDA and MAVT as a Decision Support Tool

Multi-Criteria Decision Analysis (MCDA) provides a structured, transparent framework for addressing complex decision problems where trade-offs between competing objectives are inevitable. Common to any decision making process is a systematic process that involves: defining the decision problem, identifying relevant objectives, establishing alternative decision options, and quantifying the performance of each option against these objectives.

Among these methods, Multi-Attribute Value Theory (MAVT) employs a hierarchical objective structure and value functions to convert both qualitative and quantitative data into comparable units. By assigning importance weights to each attribute, MAVT aggregates these measurements into a single composite score for each option. This approach is particularly effective for ranking alternatives and providing decision-makers with a comprehensive comparison of available options.

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1 | Literature Review and Objectives

1.1. Literature Review

A review of relevant literature reveals numerous applications of MCDA methods across diverse fields. Focusing specifically on studies combining MCDA with nuclear power decision-making, several distinct approaches emerge:

Site selection studies represent the most common application, typically employing GIS-based software to evaluate potential locations for new nuclear installations within limited geographical regions. These analyses frequently incorporate quantitative non-technical factors such as population proximity [**DAMOOM2019110**] [**BASEGMEZ2025105897**]. Abdullah et al. (2025) extend this approach by including social dimensions like public acceptance and tourism impacts, which present challenges for quantification [**ABDULLAH202510562**
Notably, Devanand et al. (2019) develop an algebraic optimization model to identify optimal SMR locations in Singapore, though like most site-focused studies, they favor a design without sistematic comparison [**DEVANAND2019339**].

A second category compares different nuclear technologies. Kiser and Otero (2023) evaluates PWRs, SMRs, and Molten Salt Reactors [**KISER2023104647**], while Locatelli and Mancini (2011) compares large and small reactor designs [**Locatelli01092012**]. Birikorang's study stands out for its regional focus on South Africa and comparison of three specific reactor models (NuScale SMR, EM2-HTGR, FTR-MSR), though its technological comparisons mix designs at different maturity levels [**BIRIKORANG2026111837**]. Notably, all three studies utilize the Analytic Hierarchy Process (AHP) or its Fuzzy variant. The last one (Birikorang) draws on the IAEA's INPRO methodology for advanced nuclear technology comparison [**IAEA_INPRO_2019**].

Broader energy comparison studies by Hernández-Torres (2025), Pasaoglu (2018), and Atilgan (2016) evaluate multiple generation options (typically eight alternatives) employing AHP-TOPSIS, AHP, and MAVT methodologies respectively. These studies use generic, averaged parameters for each technology class [**HERNANDEZTORRES2025122481**, **PASAOGLU2018654**, **ATILGAN2016168**]. In contrast, Castelazo (2014) adopts a more precise approach using Multi-Attribute Value Theory (MAVT) to compare thirteen energy technologies, employing specific reference plants (e.g. EPR reactor for nuclear power) rather than generalized technology averages [**SANTOYOCASTELAZO2014119**].

Three studies show particular relevance to the current research. Zolkaffly et al. (2014) evaluates nuclear plant models in Malaysia but suffers from outdated data and neglects non-technical aspects [**10.1063/1.4866099**]. Topaloğlu (2025) applies the Analytic Network Process (ANP) to compare reactor types (PWR, BWR, PHWR) and locations, though social considerations remain limited to site-specific factors [**TOPALOGLU2025103228**]. Goushis (2025) employs TOPSIS to assess SMR deployment in the EU for coal replacement, but focuses on a single reactor model rather than comparative technology evaluation [**GOUSHIS2025126565**].

The literature review reveals three gaps in nuclear energy decision-making research. First, most studies focus solely on site selection rather than comparing different nuclear technologies. Second, comparisons between nuclear designs lack considerration of a specific decision-making context. Third, studies that do consider specific decision-making context, examine various energy generation options as broad technology categories (e.g., nuclear, solar, wind) rather than conducting detailed analyses of specific reactor models or power plant designs.

The literature review identifies three significant research gaps in nuclear energy decision-making:

- The majority of studies focus exclusively on site selection.
- When nuclear designs are compared, the specific decision-making context in which such choices would occur is typically neglected.
- Studies that do consider decision-making contexts generally evaluate broad energy technology categories (e.g., nuclear, solar, oil) without examining specific models or power plant designs.

This study addresses these research gaps by applying Multi-Criteria Decision Analysis (MCDA), specifically employing Multi-Attribute Value Theory (MAVT), to compare reactor designs. The analysis explicitly incorporates country-specific conditions and examines their influence on the ranking of different nuclear installations.

1.2. Study Objectives

BOUNDARIES OF THE STUDY AND RESEARCH QUESTIONS - will have the sign-posts so has to be reviewd at the end

This study will focus on a limited set of reactor designs, which consideration was deemed of reasonable within the European context, considering their current state of technological advancement. In addition, four European countries have been chosen for comparison to observe how different energy markets can influence the ranking of reactor designs. The first section of the work is dedicated to the considerations that lead to these selections.

The second section is focused on the characterization based on attributes, considering country weights, of each design, their reasoning, and comments on their interactions.

Finally, we apply MAVT and comment on the resulting ranking.

2 | Preparation

2.1. Criteria for Reactor Selection in Comparative Analysis

This study focuses on a curated set of reactor designs at an advanced stage of development. The selection process begins with the the IAEA "Nuclear Reactors in the World" [28], which catalogs operational, under-construction, and decommissioned reactors globally.

The initial selection process applied two primary exclusion criteria. First, designs were eliminated based on supplier origin, removing all reactors associated with countries unlikely to engage in nuclear technology transfer with Europe under the current geopolitical climate. This filter specifically excluded reactors developed in or supplied by Russia, China, and other nations maintaining restricted technology-sharing agreements with Western markets.

Second, reactors commissioned before 2000 were excluded from consideration, as their designs represent outdated technological standards. The complete filtering methodology is documented in the accompanying analysis notebook (PRIS Data Analysis.ipynb), which details how the initial dataset was reduced to nine distinct reactor models.

From this pre-selected group, three designs - the EPR-1750, APR-1400, and AP-1000 - were prioritized for comprehensive analysis. It should be noted that while the EPR (1650 MWe) and EPR-1750 share significant design similarities, they are treated as separate models in this study. Additionally, the Advanced Boiling Water Reactor (ABWR), though listed as operational in the IAEA ARIS database, was excluded following verification through the PRIS database, which indicates all ABWR units have suspended operation since 2011 [20]. The selection process and its outcomes are summarized in Table 2.1.

Subsequently, the IAEA Advanced Reactors Information System (ARIS) database [1], was consulted to identify advanced reactor models classified as under design, under regulatory

Reactor Name	Country	Technology-sharing	Commercial Operation	Up to Date	Decision
APR1400	S.Korea	✓	✓	✓	**Selected**
CAREM Prototyp	Argentina	✓	✗	✓	Rejected
PRE KONVOI	Germany	✓	✓	✗	Rejected
EPR 1750	France	✓	✓	✓	**Selected**
AP1000	USA	✓	✓	✓	**Selected**
ABWR	Japan	✓	✗	✓	Rejected
APWR	Japan	✓	✗	✓	Rejected
EPR	France	✓	✓	✗	Rejected
OPR-1000	S.Korea	✓	✓	✗	Rejected

Table 2.1: Selection of reactor models for analysis

review, under construction, or operational. Demonstrations units and prototypes were excluded from this set, yielding 10 candidate designs. To avoid redundancy, models already included in the prior selection (e.g., Generation III+ designs captured in both datasets) were removed. Each remaining design was then assessed for development status, technological readiness, and applicability to European energy systems as summarized in Table 2.2.

So we are left with 3 Gen III+ designs and 3 advanced designs to analyze further.

Maybe here add a paragraph for each design to explain their main features.

2.2. Country Selection for Comparative Analysis

This study evaluates four European countries: France, Italy, Poland and Switzerland, to assess how national energy contexts influence the ranking of nuclear reactor designs.

In Italy, the energy market is divided into different zones, each with its own demand patterns. For this analysis, Northern Italy serves as the focal region being the country's most industrialized and energy-intensive area[4]. It is characterized by exceptionally high electricity demand, a strong reliance on imports, and above-average greenhouse gas emission intensity, making it a critical bottleneck in Italy's decarbonization efforts. Additionally, the country's historical political volatility introduces uncertainty into long-term energy planning.

France, with $\sim 70\%$ of its electricity generated by nuclear power[4], represents a mature nuclear market. Here, the marginal gains from additional reactors are likely minimal, offering insight into how reactor rankings shift in a saturated nuclear energy landscape.

Poland, heavily dependent on coal and among Europe's highest GHG emitters [4], presents an opposite case. As the country seeks to modernize its grid and reduce emissions through nuclear energy [4], it provides an opportunity to assess which reactor designs could best support its transition while meeting growing energy demand.

Finally, Switzerland provides a unique case due to its low electricity demand, hydropower-dominated grid, and prudent nuclear policy. Its constrained grid capacity and emphasis on energy independence make it ideal for assessing small modular reactors (SMRs) and their role in complementing renewables.

Reactor Name	Country	Used	Reasoning
APR+	S.Korea	✗	As reported in the manufacturer website [9] this design is mean to be an "indigenous reactor cleared of obstacles to export"
ATMEA	Japan, France	✗	Joint venture was abandoned in 2019 [4]
BWRX300	USA, Canada	✓	Design from GE-Hitachi, a lot of support from US and Canada as well as interes in the EU [25] [26].
i-SMR	S.Korea	✗	Never expressed interest in development outside south Asia and the middle east, no updates since early 2024.
iMSR400	US	✗	No interest overseas, project seems to be at an early stage of development.
Natrium	US	✗	Demo plant [45].
Nuward	France	✗	After the change in direction of 2024 the project has been set back [4].
Nuscale	US	✓	Even if there was some financial controversy in 2024 [4]. the project seems to be still moving forward in terms of goverment founding [4] and regulatory review [4].
PRISM	US	✗	GE-Hitachi considers it a "concept ... currently being put into practice in two reactors: Natrium and ARC-100" [4].
Rolls Royce SMR	UK	✓	It's moving forward and from the technological point of view this is the simples as it it litteraly a scaled down PWR[4].
SMART	S.Korea, Saudi Arabia	✗	Plan to export only to south-east asia and middle east [27].

Table 2.2: Selection of advanced reactor models for analysis

3 | Analysis

4 | Results

Bibliography

[1] International Atomic Energy Agency. *Advanced Reactors Information System (ARIS)*. Accessed: 2025-06-10. 2025. URL: https://aris.iaea.org/.

[2] AREVA NP, EDF Energy. *Sub-Chapter 15.1: Level 1 Probabilistic Safety Assessment*. Technical Report. Accessed: 2025-09-30. EDF Energy, Nov. 2012. URL: https://www.edfenergy.com/sites/default/files/sub-chapter_15.1_-_level_1_psa.pdf.

[3] M. Arslan et al. "MOX recycling in GEN 3 + EPR Reactor homogeneous and stable full MOX core". In: American Nuclear Society, 555 North Kensington Avenue, La Grange Park, IL 60526 (United States). American Nuclear Society - ANS; La Grange Park (United States), July 2013. URL: https://www.osti.gov/biblio/22257897.

[4] A. N. Author. *Example Reference Title*. Accessed: 2025-09-10. City, 2025. URL: https://www.example.com.

[5] Canadian Nuclear Safety Commission. *BWRX-300 Design: Non-Proprietary Information on Fuel Design Qualification and BWR Fuel Licensing*. Technical Report. Accessed: 2025-09-30. Canadian Nuclear Safety Commission, 2024. URL: https://api.cnsc-ccsn.gc.ca/dms/digital-medias/CMD24-H3-Ref-BWRX-300-DNNP-GNF2-Fuel-Design-Qualification-and-BWR-Fuel-Licensing-Non-Proprietary-Information.PDF.

[6] Travis S. Carless, W. Michael Griffin, and Paul S. Fischbeck. "The environmental competitiveness of small modular reactors: A life cycle study". In: *Energy* 114 (2016), pp. 84–99. ISSN: 0360-5442. DOI: https://doi.org/10.1016/j.energy.2016.07.111. URL: https://www.sciencedirect.com/science/article/pii/S0360544216310350.

[7] Electricity Maps. *2024 Carbon Intensity Data for Poland, Switzerland, France, and Northern Italy*. Accessed: 2025-09-30; Data version: January 27, 2025. Electricity Maps, Jan. 2025. URL: https://www.electricitymaps.com.

[8] ENCO. *Analysis of Small Modular Reactor (SMR) Concepts*. Technical Report. Accessed: 2025-09-30. Federal Ministry of Labour and Economy (BMLUK), Austria,

2022. URL: https://www.bmluk.gv.at/dam/jcr:26805e08-47f2-402e-8121-921e1f121144/Analysis_of_SMR_Concepts.pdf.

[9] KEPCO Engineering and Inc. Construction Company. *Reactor Descriptions.* Accessed: 2025-06-10. 2025. URL: https://www.kepco-enc.com/menu.es?mid=a20202030100.

[10] European Commission. *2050 long-term strategy.* https://climate.ec.europa.eu/eu-action/climate-strategies-targets/2050-long-term-strategy_en. Accessed: 2025-10-01. 2024. URL: https://climate.ec.europa.eu/eu-action/climate-strategies-targets/2050-long-term-strategy_en.

[11] European Commission. *Commission Delegated Regulation (EU) 2021/2139 of 4 June 2021 supplementing Regulation (EU) 2020/852 of the European Parliament and of the Council by establishing the technical screening criteria for determining the conditions under which an economic activity qualifies as contributing substantially to climate change mitigation or climate change adaptation and for determining whether that economic activity causes no significant harm to any of the other environmental objectives.* Accessed: 2025-10-01. European Union, 2025. URL: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:02021R2139-20250108.

[12] European Network of Transmission System Operators for Electricity (ENTSO-E). *Transparency Platform.* Accessed: 2025-09-30. ENTSO-E, 2025. URL: https://transparency.entsoe.eu/.

[13] European Nuclear Society (Euronuclear). *Opinion Poll: The Majority of the Swiss Support Nuclear Energy.* Accessed: 2025-09-30. European Nuclear Society, 2023. URL: https://www.euronuclear.org/news/opinion-poll-the-majority-of-the-swiss-support-nuclear-energy/.

[14] Robert J. Fetterman. "AP1000 core design with MOX loading". In: *Annals of Nuclear Energy* 36.3 (2009). PHYSOR 2008, pp. 324–330. ISSN: 0306-4549. DOI: https://doi.org/10.1016/j.anucene.2008.11.022. URL: https://www.sciencedirect.com/science/article/pii/S0306454908002983.

[15] Framatome. *GNF2 Fuel Assembly: Product Sheet.* Accessed: 2025-09-30. Framatome. May 2018. URL: https://www.framatome.com/solutions-portfolio/docs/default-source/default-document-library/product-sheets/a0529-p-us-g-en-328-05-18-htp.pdf?Status=Master&sfvrsn=60b7ce1d_2.

[16] GE Vernova. *Nuclear Energy Solutions.* Accessed: 2025-09-30. GE Vernova, 2025. URL: https://www.gevernova.com/nuclear.

[17] IFOP. *Les Français et le nucléaire : adhésion et traits d'image*. Accessed: 2025-09-30. IFOP, Sept. 2022. URL: https://www.ifop.com/article/les-francais-et-le-nucleaire-adhesion-et-traits-dimage/.

[18] Il Sole 24 Ore. "Sustainable Nuclear Comes into Law: Here Is the Government Strategy in Ten Questions and Answers". In: (2024). Accessed: 2025-09-30. URL: https://en.ilsole24ore.com/art/sustainable-nuclear-comes-law-here-is-government-strategy-ten-questions-and-answers-AGvZZsDD.

[19] International Atomic Energy Agency (IAEA). *Climate Change and Nuclear Power 2022*. Report. Accessed: 2025-09-30. International Atomic Energy Agency, 2022. URL: https://www.iaea.org/topics/nuclear-power-and-climate-change/climate-change-and-nuclear-power-2022.

[20] International Atomic Energy Agency (IAEA). *Power Reactor Information System (PRIS) Database*. Accessed: 2025-09-30. International Atomic Energy Agency, 2025. URL: https://pris.iaea.org/PRIS/home.aspx.

[21] Korea Hydro and Nuclear Power Co., Ltd. (KHNP). *Major Characteristics of the EU-APR Reactor*. Presentation. Accessed: 2025-09-30. Apr. 2018. URL: https://sep.com.pl/photo/files/07b%20-%20KHNP%20-%20Major%20Characteristics%20of%20EU-APR%20Reactor.pdf.

[22] Bojie Liu et al. "Critical review of nuclear power plant carbon emissions". In: *Frontiers in Energy Research* Volume 11 - 2023 (2023). ISSN: 2296-598X. DOI: 10.3389/fenrg.2023.1147016. URL: https://www.frontiersin.org/journals/energy-research/articles/10.3389/fenrg.2023.1147016.

[23] NEA. *Unlocking Reductions in the Construction Costs of Nuclear*. Accessed: 2025-09-10. Paris, 2020. URL: https://www.oecd-nea.org/jcms/pl_30653/unlocking-reductions-in-the-construction-costs-of-nuclear.

[24] NEA. *Unlocking Reductions in the Construction Costs of Nuclear*. Paris: OECD Publishing, 2020.

[25] World Nuclear News. *Early works agreement for BWRX-300 SMRs in Finland and Sweden*. Accessed: 2025-06-10. 2025. URL: https://www.world-nuclear-news.org/articles/early-works-agreement-for-bwrx-300-smrs-in-finland-and-sweden.

[26] World Nuclear News. *Hungary looking at deployment of US small modular reactors*. Accessed: 2025-06-10. 2025. URL: https://www.world-nuclear-news.org/articles/hungary-looking-at-deployment-of-us-small-modular-reactors.

[27] World Nuclear News. *South Korean SMR design approved by regulator*. Accessed: 2025-06-10. Sept. 26, 2024. URL: https://www.world-nuclear-news.org/articles/south-korean-smr-design-approved-by-regulator.

[28] *Nuclear Power Reactors in the World.* entity 2. Vienna: INTERNATIONAL ATOMIC ENERGY AGENCY, 2024. ISBN: 978-92-0-122224-4. URL: /publications/15748/nuclear-power-reactors-in-the-world.

[29] NucNet. "Majority Continues to Support Nuclear Energy, Says Survey". In: (Mar. 2023). Accessed: 2025-09-30. URL: https://www.nucnet.org/news/majority-continues-to-support-nuclear-energy-says-survey-3-1-2023.

[30] NucNet. "Poland Survey". In: (Dec. 2024). Accessed: 2025-09-30. URL: https://www.nucnet.org/news/more-than-92-support-plans-for-first-nuclear-power-plant-12-5-2024.

[31] NuScale Power. *NuScale Announces MOX Capability.* Accessed: 2025-09-30. NuScale Power, Jan. 2016. URL: https://www.nuscalepower.com/press-releases/2016/nuscale-announces-mox-capability.

[32] NuScale Power. *NuScale Power Announces an Additional 25 Percent Increase in NuScale Power Module Output.* Accessed: 2025-10-01. Nov. 2020. URL: https://www.nuscalepower.com/press-releases/2020/nuscale-power-announces-an-additional-25-percent-in-nuscale-power-module-output.

[33] NuScale Power. *NuScale Power Module: Product Overview.* Accessed: 2025-09-30. NuScale Power, 2025. URL: https://www.nuscalepower.com/products/nuscale-power-module.

[34] OECD Nuclear Energy Agency (NEA). *Technical and Economic Aspects of Load Following with Nuclear Power Plants.* Technical Report. Accessed: 2025-09-30. OECD Nuclear Energy Agency, June 2011. URL: https://www.oecd-nea.org/upload/docs/application/pdf/2021-12/technical_and_economic_aspects_of_load_following_with_nuclear_power_plants.pdf.

[35] Our World in Data. *Death Rates from Energy Production per TWh.* Accessed: 2025-09-30; Data sources: Markandya & Wilkinson (2007); Sovacool et al. (2016); UNSCEAR (2008 & 2018). Our World in Data, 2025. URL: https://ourworldindata.org/grapher/death-rates-from-energy-production-per-twh.

[36] H. Paillere. "Nuclear energy for a net zero world". In: (2021), [12 p.]

[37] Power Technology. "Rolls Royce plans first UK modular nuclear reactor for 2029". In: *Power Technology* (Apr. 2022). Accessed: 2025-10-01. URL: https://www.power-technology.com/news/uk-first-smr-rolls-royce/.

[38] Reuters Events. "GE Hitachi chases gas plant displacement with new 300 MW reactor". In: *Reuters Events / Nuclear* (Apr. 2018). Accessed: 2025-10-01. URL: https://www.reutersevents.com/nuclear/ge-hitachi-chases-gas-plant-displacement-new-300-mw-reactor.

[39] Rolls-Royce SMR. *Generic Design Assessment (GDA) Public Consultation: Have Your Say*. Accessed: 2025-09-30. Rolls-Royce SMR, 2025. URL: `https://gda.rolls-royce-smr.com/have-your-say`.

[40] Rolls-Royce SMR. *Our Approach to Small Modular Reactors*. Accessed: 2025-09-30. Rolls-Royce SMR, 2025. URL: `https://www.rolls-royce-smr.com/our-approach`.

[41] Rolls-Royce SMR. *Rolls-Royce SMR Select UK-Based Fuel Solution*. Accessed: 2025-09-30. Rolls-Royce SMR, Oct. 2023. URL: `https://www.rolls-royce-smr.com/press/rolls-royce-smr-select-uk-based-fuel-solution`.

[42] Geoffrey Rothwell. "Projected electricity costs in international nuclear power markets". In: *Energy Policy* 164 (2022), p. 112905. ISSN: 0301-4215. DOI: `https://doi.org/10.1016/j.enpol.2022.112905`. URL: `https://www.sciencedirect.com/science/article/pii/S0301421522001306`.

[43] Björn Steigerwald et al. "Uncertainties in estimating production costs of future nuclear technologies: A model-based analysis of small modular reactors". In: *Energy* 281 (2023), p. 128204. ISSN: 0360-5442. DOI: `https://doi.org/10.1016/j.energy.2023.128204`. URL: `https://www.sciencedirect.com/science/article/pii/S0360544223015980`.

[44] SWI swissinfo.ch. "Risks Remain as Switzerland Weighs a Return to Nuclear Power". In: (2025). Accessed: 2025-09-30. URL: `https://www.swissinfo.ch/eng/emissions-reduction/risks-remain-as-swiss-weigh-a-return-to-nuclear-power/89549444`.

[45] TerraPower. *Natrium Reactor Technology*. Accessed: 2025-06-10. 2025. URL: `https://www.terrapower.com/natrium/`.

[46] U.S. Nuclear Regulatory Commission. *Design Description*. Technical Report. Accessed: 2025-09-30. U.S. Nuclear Regulatory Commission, 2005. URL: `https://www.nrc.gov/docs/ML0522/ML052280170.pdf`.

[47] U.S. Nuclear Regulatory Commission. *Safety Evaluation by the Office of Nuclear Reactor Regulation: NuScale Power Module*. ML23161A411. Accessed: 2025-09-30. U.S. Nuclear Regulatory Commission, June 2011. URL: `https://www.nrc.gov/docs/ML1117/ML11171A411.pdf`.

[48] U.S. Nuclear Regulatory Commission. *Standard Review Plan for Environmental Reviews for Nuclear Power Plants*. Technical Report. Accessed: 2025-09-30. U.S. Nuclear Regulatory Commission, 2017. URL: `https://www.nrc.gov/docs/ML1700/ML17007A001.pdf`.

[49] Westinghouse Electric Company. "Poland's First Nuclear Plant Moves Forward with Westinghouse-Bechtel-PEJ Agreement". In: (2025). Accessed: 2025-09-30. URL:

https://info.westinghousenuclear.com/news/polands-first-nuclear-plant-moves-forward-with-westinghouse-bechtel-pej-agreement.

[50] World Nuclear Association. *Switzerland's Nuclear Industry Profile*. Accessed: 2025-09-30. World Nuclear Association, 2025. URL: https://world-nuclear.org/information-library/country-profiles/countries-o-s/switzerland.

[51] World Nuclear News. "Go-Ahead for Preliminary Works at Polish Plant Site". In: (Sept. 2025). Accessed: 2025-09-30. URL: https://www.world-nuclear-news.org/articles/go-ahead-for-preliminary-works-at-polish-plant-site.

[52] World Nuclear News. "Poll Finds Majority of Swiss Support Nuclear New Build". In: (2023). Accessed: 2025-09-30. URL: https://www.world-nuclear-news.org/articles/poll-finds-majority-of-swiss-support-nuclear-new-build.

A | Appendix A

If you need to include an appendix to support the research in your thesis, you can place it at the end of the manuscript. An appendix contains supplementary material (figures, tables, data, codes, mathematical proofs, surveys, ...) which supplement the main results contained in the previous chapters.

B | Appendix B

It may be necessary to include another appendix to better organize the presentation of supplementary material.

List of Figures

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List of Symbols

Variable	Description	SI unit
$\boldsymbol{u}$	solid displacement	m
$\boldsymbol{u}_f$	fluid displacement	m

Acknowledgements

Here you might want to acknowledge someone.